TOPIC 10

Elastic Net Regression: Explained

The original class slide, with a plain explanation under it

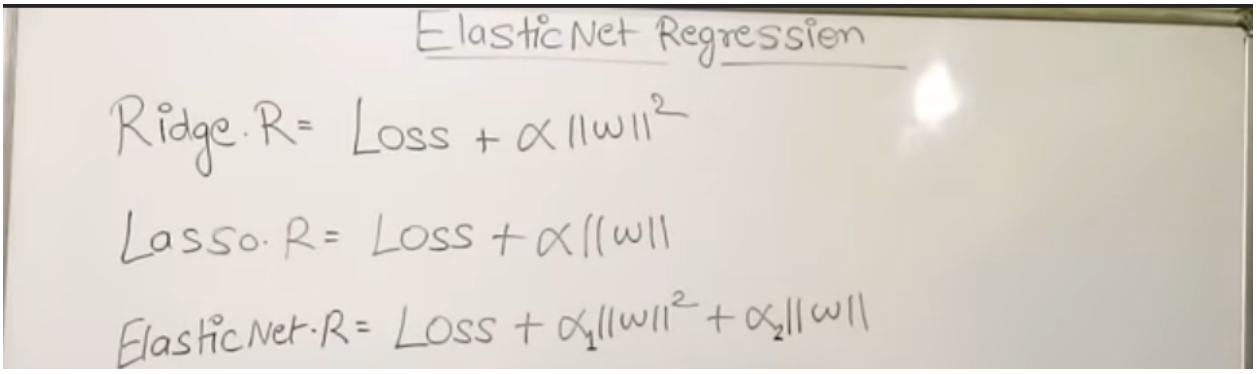
> **NOTE** **How to read this document**
>
> - Every original slide is kept exactly as it was, with its slide number under it
> - Under each slide (or small set of slides) you get a plain explanation: **what you are looking at** and **what it means**
> - The slides come from a recorded class, so a few look like screenshots of a video: that is normal

Elastic Net combines the two penalties you have already met, Ridge and Lasso, in one formula. This topic is a single whiteboard slide.

Words you will meet

Word	Plain meaning
Ridge penalty	$\alpha\|w\|^2$: the squares of the coefficients
Lasso penalty	$\alpha\|w\|$: the absolute values of the coefficients
Elastic Net	Uses both penalties, each with its own strength (α_1 and α_2)

Slide 1: Ridge, Lasso and Elastic Net on one board



Slide 1

The three formulas, one under the other:

- **Ridge:** Loss + $\alpha\|w\|^2$. It penalises the **squares** of the coefficients, so they shrink smoothly but do not reach zero.
- **Lasso:** Loss + $\alpha\|w\|$. It penalises the **absolute values**, so some coefficients become exactly zero (features removed).
- **Elastic Net:** Loss + $\alpha_1\|w\|^2 + \alpha_2\|w\|$. It simply **adds both penalties**, with their own strengths α_1 and α_2. So it shrinks coefficients like Ridge **and** can drop features like Lasso.

$$\text{ElasticNet loss} = \text{Loss} + \alpha_1\|w\|^2 + \alpha_2\|w\|$$